

Subject Description Form

Subject Code	EE4019 / EE4019B
Subject Title	Intelligent Transportation Systems
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Pre-requisite for EE4019: EE2029 or EEE2003 Pre-requisite for EE4019B: EE2029B
Objectives	1. To explore advanced transportation technologies and demonstrate their real-world applications. 2. To provide a comprehensive understanding of the challenges faced in transport operations, and the corresponding technological solutions required to address them. 3. To enable the evaluation of appropriate methodologies and develop an awareness of the design and implementation considerations for advanced transportation technologies.
Subject Intended Learning Outcomes	Upon completion of the subject, students should be able to: a. Demonstrate a comprehensive understanding of the key issues and challenges faced in transport operations. b. Explain how information and communications technologies are leveraged to address various transport-related challenges. c. Recognise and identify the fundamental design considerations in the development of intelligent transport systems.
Subject Synopsis/ Indicative Syllabus	1. Data Collection and Data Analytics: Introduction to the collection and analysis of transportation data, data sources, and big data analytics for optimising transportation systems. 2. Traveller Information Systems: Understanding the benefits of providing information to travellers, including estimating and predicting travel times, and real-time passenger information. 3. Traffic management with ITS: Applications of ITS in managing traffic on motorways and arterial roads, such as ramp metering, variable speed limits, lane usage management, electronic toll collection, public transport priority and incident detection. 4. Green transportation systems: Explore the challenges of operating electric and alternative fuels vehicles including scheduling EV charging, supply of alternative fuels and charging infrastructure. 5. Artificial Intelligence (AI) applications in traffic management: Applications of the latest advancements in using machine learning to predict traffic patterns and reinforcement learning to control traffic. 6. Connected Autonomous Vehicles and Cooperative ITS: The future of transportation with connected autonomous vehicles and the use of vehicle-to-

	vehicle, vehicle-to-infrastructure, and vehicle-to-everything (V2X) communication to enhance safety and efficiency.				
Teaching/Learning Methodology	Delivery of the subject is mainly through formal lectures and complemented by tutorials. Assignment provides students hands-on experience in processing and analysing big-data, while report-writing enables students to practise writing skill.				
	Teaching/Learning Methodology	Outcomes			
		a	b	c	
	Lectures	✓	✓	✓	
	Tutorials	✓	✓	✓	
	Assignment			✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	1. Mini Projects	40%	✓	✓	✓
	2. Term Assessment	40%	✓	✓	✓
	3. Assignment	20%			✓
	Total	100%			
	The mini projects allow assessment of outcomes covering principles, techniques, design and application, supplemented by continuous assessments. Assignments enable students to explore and apply data analytics and evaluate transport system performance.				
Student Study Effort Expected	Class contact:				
	▪ Lecture/Tutorial		39 Hrs.		
	Other student study effort:				
	▪ Mini projects and assignment		30 Hrs.		
	▪ Self-study		38 Hrs.		
	Total student study effort		107 Hrs.		
Reading List and References	Reference books:				
	1. US DoT, ITS ePrimer, ITS Joint Program Office, www.pcb.its.dot.gov/eprimer/				
	2. PIARC, Cooperative Vehicle Highway Systems, Technical Committee 2.1 Road Network Operations, 2016.				
	3. R. Gordon, Intelligent Transportation Systems: Functional Design for Effective Traffic Management, Springer, 2016.				